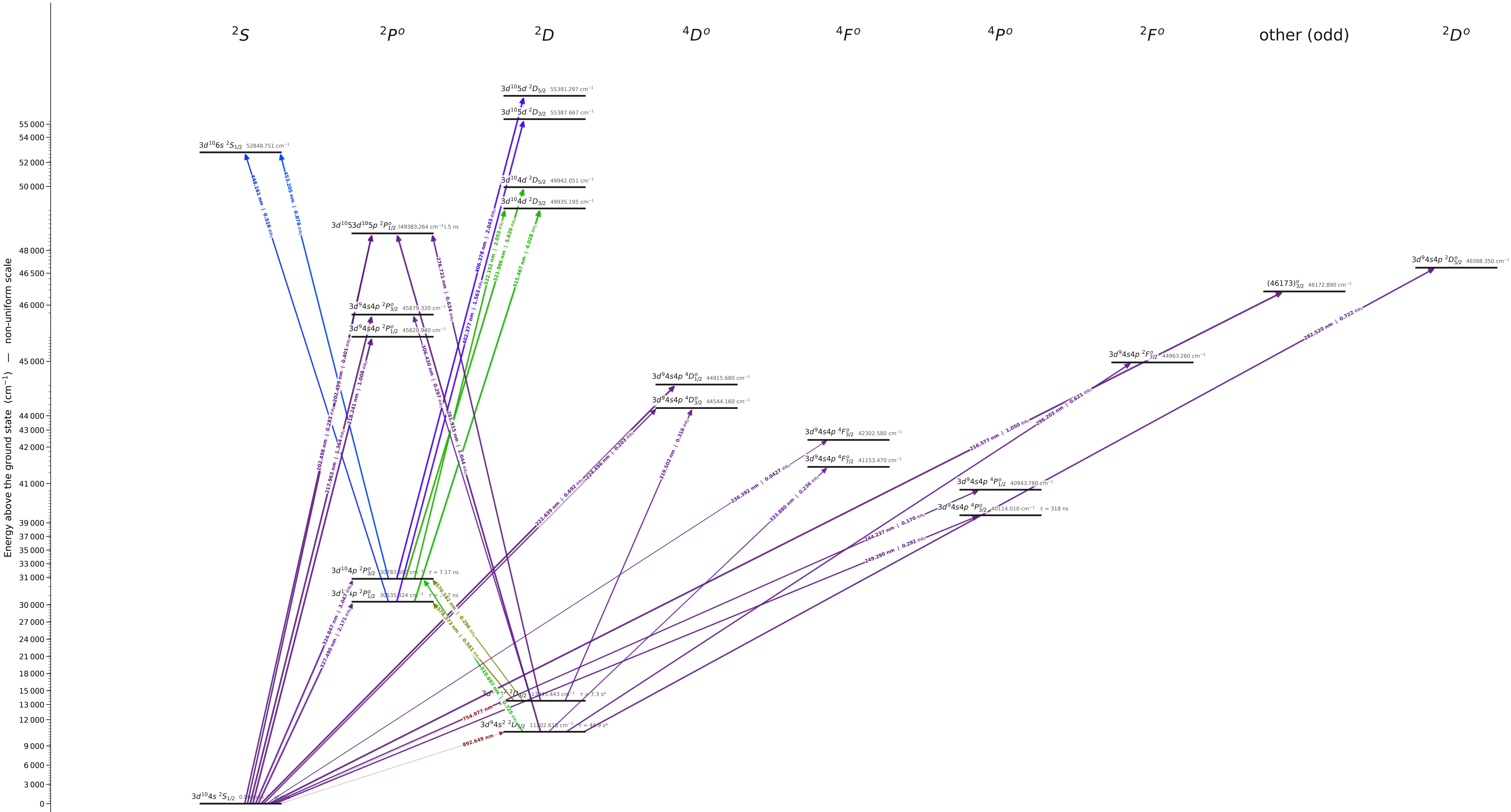
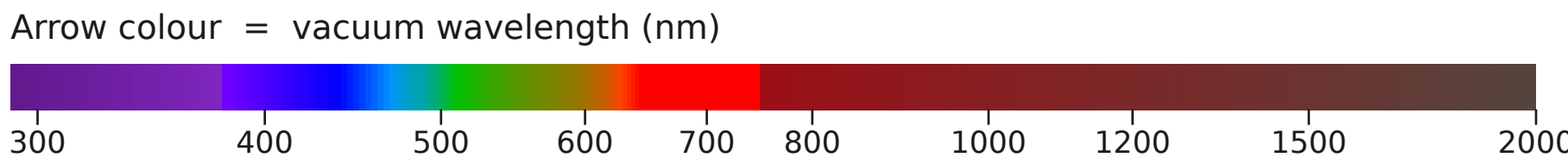


Cu^+ ionisation limit $62317.46 \text{ cm}^{-1} = 7.72638 \text{ eV} (160.469 \text{ nm})$

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle U \text{er} U\rangle $ (eao)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$3d^{10}4s\ ^2S_{1/2} - 3d^{10}5p\ ^2P_{3/2}$	202.4991	202.4339	1480.4634	0.401	9.800e+6	6.241	25	202.4 nm line used for pulsed level-crossing spectroscopy...
$3d^{10}4s\ ^2S_{1/2} - 3d^94s4p\ ^4P_{9/2}^o$	244.2373	244.1633	1227.4636	0.170	2.010e+6	-	25	244.2 nm line used for in-gas-cell resonance-ionisation s...
$3d^{10}4s\ ^2S_{1/2} - 3d^94s4p\ ^4P_{3/2}^o$	249.2895	249.2143	1202.5878	0.292	2.790e+6	0.5005	88.7	249.2 nm line used for resonance-ionisation spectroscopy...
$3d^{10}4s\ ^2S_{1/2} - 3d^{10}4p\ ^2P_{3/2}^o$	324.8473	324.7536	922.8270	3.047	1.372e+8	22.2	98.4	D2 resonance line 324.8 nm
$3d^{10}4s\ ^2S_{1/2} - 3d^{10}4p\ ^2P_{1/2}^o$	327.4896	327.3952	915.4260	2.171	1.359e+8	21.89	98.8	D1 resonance line 327.4 nm
$3d^94s^2\ ^2D_{5/2} - 3d^{10}4p\ ^2P_{3/2}^o$	510.6971	510.5548	587.0260	0.725	2.000e+6	22.2	1.43	510.6 nm green copper-vapour-laser line
$3d^94s^2\ ^2D_{3/2} - 3d^{10}4p\ ^2P_{3/2}^o$	578.3730	578.2126	518.3376	0.561	1.650e+6	21.89	1.2	578.2 nm yellow copper-vapour-laser line
$3d^{10}4s\ ^2S_{1/2} - 3d^94s^2\ ^2D_{5/2}$	754.9766	754.7688	397.0884	-	6.270e-07	2.18e-08	0.000458	second proposed clock transition
$3d^{10}4s\ ^2S_{1/2} - 3d^94s^2\ ^2D_{3/2}$	892.6485	892.4035	335.8460	-	2.230e-02	3.545e-09	100	E2 clock transition proposed for a Cu lattice clock

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
$3d^{10}4s\ ^2S_{1/2}$	5866.915(5)	-	F=1: -7333.64	F=2: +4400.19		
$3d^94s^2\ ^2D_{5/2}$	749.100	186.000	F=1: -3802.57	F=2: -2453.18	F=3: -289.57	F=4: +2855.62
$3d^94s^2\ ^2D_{3/2}$	1851.000	137.400	F=0: -6769.50	F=1: -5055.90	F=2: -1491.30	F=3: +4199.10
$3d^{10}4p\ ^2P_{3/2}$	506.600	-	F=1: -633.25	F=2: +379.95		
$3d^{10}4p\ ^2P_{1/2}$	194.5(11)	-28.0(6)	F=1: -764.38	F=2: -541.88	F=2: -124.88	F=3: +430.62
$3d^94s4p\ ^4P_{3/2}$	2129.000	-	F=0: -7983.75	F=1: -5854.75	F=2: -1596.75	F=3: +4790.25
$3d^94s4p\ ^4P_{1/2}$	2432(8)	-	F=1: -3040.00	F=2: +1824.00		
$3d^94s4p\ ^4F_{7/2}$	1139.000	-	F=2: -7688.25	F=3: -4271.25	F=4: +284.75	F=5: +5979.75
$3d^94s4p\ ^4F_{5/2}$	305.000	98.000	F=0: -1021.25	F=1: -814.25	F=2: -302.25	F=3: +710.75
$3d^94s4p\ ^4D_{3/2}$	124.000	-3.000	F=0: -468.75	F=1: -341.75	F=2: -90.75	F=3: +278.25
$3d^94s4p\ ^4D_{1/2}$	-510.000	-	F=1: +637.50	F=2: -382.50		
$3d^94s4p\ ^2F_{7/2}$	1919.000	-	F=2: -12953.25	F=3: -7196.25	F=4: +479.75	F=5: +10074.75
$3d^94s4p\ ^2P_{3/2}$	948.000	-	F=1: -1185.00	F=2: +711.00		
$3d^94s4p\ ^2P_{1/2}$	1619.000	27.000	F=0: -6037.50	F=1: -4445.50	F=2: -1234.50	F=3: +3649.50
$(46173)3s_{1/2}^0$	2100.000	-	F=0: -7875.00	F=1: -5775.00	F=2: -1575.00	F=3: +4725.00
$3d^94s4p\ ^2D_{5/2}$	2533.000	-	F=1: -13298.25	F=2: -8322.25	F=3: -633.25	F=4: +9498.75
$3d^{10}5s\ ^2P_{3/2}$	61.7(9)	4.9(7)	F=0: -225.25	F=1: -168.45	F=2: -49.95	F=3: +140.05
$3d^{10}5p\ ^2P_{3/2}^o$	334.600	-	F=1: -418.25	F=2: +250.95		
$3d^{10}4d\ ^2D_{3/2}$	67.000	-	F=0: -251.25	F=1: -184.25	F=2: -50.25	F=3: +150.75
$3d^{10}4d\ ^2D_{5/2}$	111.000	-	F=1: -582.75	F=2: -360.75	F=3: -27.75	F=4: +416.25

Measured A, B constants; sources in the data file.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$3d^{10}4s^2S_{1/2} - 3d^94s4p^4P_{9/2}$	249.290	63-65	1055.4(26)
$3d^94s^2D_{5/2} - 3d^{10}4p^2P_{3/2}$	510.697	63-65	-2128.00
$3d^{10}4p^2P_{3/2} - 3d^{10}4d^2D_{3/2}$	1515.647	63-65	363.00
$3d^{10}4p^2P_{3/2} - 3d^{10}4d^2D_{5/2}$	521.966	63-65	354.00

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$